

Dr. ABHIJIT LINCON

Physician Researcher | Biochemical sensing | Molecular Biophysics | Bio-electrochemistry

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Summary

Physician-researcher with a strong foundation in clinical medicine and interdisciplinary research, specializing in protein biophysics, biochemical and bioelectrical sensing, and biomolecular diagnostics. My work focuses on understanding electrical events in protein unfolding and aggregation, integrating electrochemical sensing, computational modeling, and experimental design to develop innovative diagnostic and drug-testing platforms. Driven by a passion for discovery and translational impact, I aim to bridge fundamental science with real-world biomedical applications.

Education

Doctor of Philosophy (Ph.D.)

Indian Institute of Technology, Kharagpur, India 2017-2024

Thesis Title: Unraveling Structural Dynamics in Unfolding of Protein by non-Faradic electrical impedance spectroscopy **CGPA:** 9.75/10

Supervisors: Prof. Soumen Das & Prof. Sunando DasGupta

Master in Medical Science & Technology (MMST)

Indian Institute of Technology, Kharagpur, India 2014-2017

Thesis Title: Development of Paper-Based Quantitative Detection System for Serum Lipase

Supervisor: Prof. Soumen Das

CGPA: 9.58/10

Bachelor of Medicine, Bachelor of Surgery (MBBS)

SCB Medical College, Cuttack, India 2006-2012

CGPA/Marks: 61.51% – First Class with distinction

- ◆ Acquired a deep understanding of human biology, disease mechanisms, and therapeutic principles
- ◆ Exposure to early-stage biomedical research and clinical diagnostics
- ◆ Integrated clinical decision-making with evidence-based practices

Higher Secondary (CBSE)

DAV Public School, Kalinga Area, Talcher, India 2003-2005

CGPA/Marks: 92% – 7th Rank in Odisha (Physics, Chemistry, Mathematics, Biology, English)

Secondary (BSE, Odisha) 2000-2003

Colliery High School, Dera, Talcher, India

CGPA/Marks: 86.31%

Subjects: Mathematics, Science, Odia, English, Sanskrit, Social science

Experience

Senior Project Associate 2024-2025

Biosensors & Microfluidics Lab

Indian Institute of Technology, Kharagpur, India

Molecular dynamics simulations using GROMACS on the IIT Kharagpur HPC cluster to investigate the thermal unfolding of porcine pancreatic lipase.

Research Interest

Bio-electrochemical medical diagnostics | Biosensors | Biochemistry | Molecular Biology | Cellular Biophysics | Protein-aggregation | Nanomaterials | Nano-medicine | Biomicrofluidics | Signal processing | Science of small



Key Achievements

Institute Silver Medal, IIT Kharagpur

Awarded to be the best student in order of merit among the graduating MMST

Illuminati Hackathon, IIT Kharagpur, Runner up. (2016)

Development of Real-time ophthalmic squint detection application

Young Student Innovator Fellowship, SIEMENS Healthcare, Goa. (2015)

First Rank in MMST-GATE 2014

Qualifying Examination For Medical Doctors to Study (2014) Masters in Medical Science and Technology.

All India rank 1425 in the Premedical examination, ODISHA JEE Rank 122 (Medical) and 286 (Engineering) (2006)

7th Rank in AISSCE (CBSE) (2005)



Skills

❖ Project Management

❖ Leadership

❖ Data Analysis

❖ Effective Communication

❖ Teamwork

❖ Critical Thinking

❖ Time Management

❖ Experimental Design



Extra-academic activities

- Academic secretary, Students Council, School of Medical Science and Technology, IIT Kharagpur. (2018-2020)
- General Secretary, Medical Students Council, SCB Medical College Cuttack. (2008-2009)
- Volunteered in NCC and NSS Camps as a Medical officer
- Social Involved in Health Promotion and Education for School students in ODISHA

Senior Research Fellow

 2019-2024

Biosensors & Microfluidics Lab
Indian Institute of Technology, Kharagpur, India

- Developed a label-free drug testing platform using impedance-based microelectrodes to study protein unfolding amyloid disaggregation, and atherosclerosis
- Conducted interdisciplinary research in biophysics and biosensing, integrating GROMACS simulations, COMSOL modeling, and spectroscopic validation (CD, FTIR, DSC)
- Investigated real-time electrical and dielectric responses of proteins under thermal and electric field conditions

Medical Officer

B.C. Roy Technology Hospital Kharagpur, India
 2013-2014  West Bengal, India

- Managed outpatient and emergency care for students, staff, and dependents
- Performed primary diagnosis, and minor procedures, and coordinated specialist referrals

Medical Officer

Community Health Centre, Khajuriakata Govt. Hospital
 2013-2013  Odisha, India

- Delivered primary and emergency care, managed maternal-child health and immunization services
- Supervised Health workers; implemented national health programs (RNTCP, NCD, malaria, etc.)
- Maintained health records, conducted outreach camps, and supported public health reporting



Teaching Experience

Spring, 2019-2022 **MM69320:** Medical Electronics, IIT Kharagpur.

Fall, 2022 : **MM61317:** Biological Physics, IIT Kharagpur.



Laboratory Skill

- Electrochemical Analysis
- Immuno-assay & Biochemical Assays
- Raman/UV/FTIR/ Atomic Force Spectroscopy
- Cell culture & Microscopy (Optical & Fluorescence)
- Electron Microscopy/X-ray diffraction
- Soft Lithography
- Metal Thin-film deposition
- Nanoparticle Synthesis and Characterization
- Oxygen Plasma & Zetasizer
- Microfabrication & 3D printing



Computer Skill

- C++
- MS Excel
- Latex
- Corel Draw
- Adobe Illustrator
- Python



Simulation Tools

- COMSOL Multiphysics
- ABAQUS
- Molecular Dynamics: GROMACS



Data Analysis Tools

- MATLAB
- MS Excel
- Latex
- SPSS
- Python
- Origin Lab



Patent

Lincon, A., Yadav, S.K., DasGupta, S., Das, S. (2024) A label-free multispectral anti-atherosclerotic drug testing system. (Indian Patent filed: Patent Application No. 202431046888 dated 18-06-2024).



Publications

- 2025
- Lincon, A.**, Yadav, S. K., Das, S., Mandal, M., DasGupta, S., & Das, S. (2025). Capturing in-vitro electro-mechanochemical signals in a label-free drug testing system for atherosclerosis. *Sensors and Actuators B: Chemical*, 427, 137182.
- 2024
- Yadav, S. K., Das, S., **Lincon, A.**, Saha, S., BoseDasgupta, S., Ray, S. K., & Das, S. (2024). Gelatin-decorated Graphene oxide: A nanocarrier for delivering pH-responsive drug for improving therapeutic efficacy against atherosclerotic plaque. *International Journal of Pharmaceutics*, 651, 123737.
- 2024
- Chowdhury, P., **Lincon, A.**, Bhowmik, S., Ojha, A. K., Chaki, S., Samanta, T., ... & Dasgupta, S. (2024). Biodegradable solid polymer electrolytes from the discarded cataractous eye protein isolate. *ACS Applied Bio Materials*, 7(4), 2240-2253.
- 2023
- Lincon, A.**, Mohapatra, P., Das, S., & DasGupta, S. (2024). Probing silver nanoparticle mediated mitigation of UV-photolysis in proteins by electrical impedance analysis. *International Journal of Biological Macromolecules*, 256, 128271.

2022 **Lincon, A.**, Das, S., & DasGupta, S. (2022). Capturing protein denaturation using electrical impedance technique. *Journal of Molecular Liquids*, 360, 119301.

2021 Basu, M., Parihar, V., **Lincon, A.**, Joshi, V. P., Das, S., & DasGupta, S. (2021). Development of graphene oxide–PDMS composite dielectric for rapid droplet movement in digital microfluidic applications. *Chemical Engineering Science*, 230, 116175.



Conferences

2025 **Abhijit Lincon**, Sunando DasGupta, and Soumen Das. Integrated in-vitro Label-free Multispectral Sensing of Micropathological Fingerprints. International Conference on Frontiers in Translational Bioengineering for Healthcare 2025, Ranchi, India

2024 **Abhijit Lincon** Sandeep Kumar Yadav and Soumen Das. Graphene oxide decorated with gelatin: A nanocarrier for delivering pH-responsive therapeutics to improve its efficacy towards atherosclerosis plaque. 48th FEBS Congress, 2024, Rome, Italy.

2023 **Abhijit Lincon**, Sunando DasGupta, and Soumen Das. Sensing dielectric relaxation in thermal unfolding of lipase. In ACS Spring Meet. American Chemical Society, 2023, Indianapolis, US.

References

Referee 1: Dr. Soumen Das, Professor, School of Medical Science & Technology. Indian Institute of Technology, Kharagpur, India

Referee 2: Dr. Sunando DasGupta, Professor, Department of Chemical Engineering. Indian Institute of Technology, Kharagpur, India

Referee 3: Dr. Mahitosh Mandal, Professor, School of Medical Science & Technology. Indian Institute of Technology, Kharagpur, India

